



Sina Emami

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Working Experience

Mercedes-Benz

Stuttgart, Germany

MASTER'S THESIS – EMBEDDING MODELS FOR RETRIEVAL AND CONTRADICTION DETECTION

Oct. 2025 – Present

- **Fine-tuning** embedding models to improve retrieval quality and document-level contradiction detection within a RAG workflow.
- Generating human-seeded synthetic data using advanced reasoning models to enrich training data and stress-test contradiction detection.
- Building an **MLflow-based** training pipeline on **Databricks**, integrating experiment tracking, model versioning, and deployment-ready artifacts as part of an **MLOps** workflow.

AI ENGINEER (WORKING STUDENT)

Mar. 2025 – Sep. 2025

- Built and validated retrieval+decision pipelines with quantitative metrics (precision/recall-style, robustness, failure analysis), focusing on reliability and traceability.
- Developed classification models with human-seeded synthetic data for stress-testing; performed systematic evaluation and ablations to validate generalization.
- Performed **systematic evaluations** of the agent pipeline, including retrieval quality assessment (precision/recall-style metrics, grounding checks) and prompt evaluation (robustness, hallucination rate, response quality).
- Integrated LLMs and retrieval pipelines into internal tools, focusing on reliability and safety in generated outputs.

Technical University of Munich (TUM) Internship

Munich, Germany

INTERN - LARGE LANGUAGE MODELS AND NLP

October. 2024 - March. 2025

- **Fine-tuned** LLMs using the **QLORA** algorithm to generate news articles in varied tones and styles, employing **reverse prompt engineering** for enhanced tone optimization.
- Developed a dynamic topic modeling framework leveraging **prompt engineering** to extract key topics, followed by sentence-based mapping to generate concise, context-aware summaries.

Fraunhofer IIS

Munich, Germany

RESEARCH ASSISTANT - DATA SCIENCE AND MACHINE LEARNING

May. 2024 - February. 2025

- Implementing **semantic segmentation** tasks using **semi-supervised learning** and **active learning** to enhance model performance defect detection with less labeled data.
- Improved model robustness under domain shift (manufacturing/inspection variability) using domain adaptation/generalization; reduced labeling via active learning.
- Working on **MLOps** tasks to create a more efficient machine learning pipeline using **MLflow** and **Hydra**.

OrthoCoPilot - TUM

Munich, Germany

MACHINE LEARNING ENGINEER (APPLICATION PROJECT)

Mar. 2024 - September

- Worked on natural language processing (NLP) tasks and **large language models (LLMs)** to enhance the diagnostic process for patient care.
- Employed **prompt engineering** techniques to refine and improve LLM responses for more accurate patient diagnosis.
- Utilized **Retrieval-Augmented Generation (RAG)** to enhance the question-answering capabilities of the LLMs, improving the quality and relevance of patient diagnostics.

HypoVereinsbank - UniCredit

Munich, Germany

MACHINE LEARNING ENGINEER (WORKING STUDENT)

Sep. 2023 - Dec. 2023

- Developed an end-to-end analytical pipeline using Python (**Transformers**, **PyTorch**, **PySpark**, **NLTK**, **spaCy**) to perform sentiment analysis (91% accuracy) and topic modelling of news content, enabling faster and more precise market-trend responses.

Education

M.Sc. in Data Engineering and Analytics

Munich, Germany

TECHNICAL UNIVERSITY OF MUNICH(TUM)

Apr. 2023 - Current(Mar. 2026)

- My overall grade in the German grading system is **2.2**.
- Currently conducting my Master's thesis at Mercedes-Benz on fine-tuning embedding models to improve retrieval quality and document-level contradiction detection in retrieval-augmented generation (RAG) workflows.

B.S. in Computer Engineering

Tehran, Iran

AZAD UNIVERSITY, CENTRAL TEHRAN BRANCH(IAUCTB)

Sep. 2017 - Jul. 2021

- overall grade of **17.74/20.0**. (In German grade scale: **1.8**)

Publications

ContextBias: A Controlled Evaluation of Context Effects in Text-to-Image Models

SUBMITTED TO CVPR, UNDER REVIEW

- Authors: Sina Emami, S. Koli.

Skills

Programming Languages	Python, SQL
Machine Learning & AI	LLMs, VLMs, Diffusion Models, Computer Vision, NLP, Multi-Agent Systems, RAG, Model Evaluation
ML & AI(Tools & Libraries)	PyTorch, Hugging Face, Stable Diffusion, LangChain, LangGraph, CrewAI, Ollama, MLflow
Data Analysis & Visualization	Pandas, NumPy, Statsmodels, Matplotlib, Seaborn, Tableau
Web Development	FastAPI, Flask, Pytest, Streamlit
Database Management	Postgres, Cosmos DB, PySpark, MongoDB, Redis
Development Tools & Environments	Databricks, Palantir, Git, Linux, Docker, Jira, Agile, Confluence
Soft Skills	Problem-Solving, Communication, Continuous Learning

Projects

Bias Detection in Vision-Language Models (VLMs)

USING STABLE-DIFFUSION FOR IMAGE GENERATION & MULTI-AGENT ANALYSIS WITH CREWAI

- Generated diverse synthetic image sets using **Stable Diffusion** and **DALL-E** to probe attribute distributions (e.g., gender, ethnicity, profession) across different contexts.
- Built a **multi-agent** analysis pipeline with CrewAI, using **GPT-5-based agents** as VLM tools to produce rich image descriptions and extract sensitive attributes and contextual cues.
- Applied **statistical analysis** to attribute repetition and co-occurrence frequencies to quantify bias and assess the fairness of VLM behaviour across scenarios.
- Proposed mitigation strategies and evaluation protocols aligned with ethical AI and model safety considerations.

Covid-19 lung CT Scan segmentation

THIS MODEL IS TRAINED TO SEGMENT PARTS INFECTED BY COVID-19 IN CT SCAN

- The dataset from [Kaggle page](#) is used.
- Used Python, Tensorflow, Keras, OpenCV, Numpy, Pandas, Matplotlib
- **Histogram Equalization** is used to make infection parts more visible and augmentation is used to create more data for training better.
- The test set got the **93% for AUC, 0.93% sensitivity**, and **0.99% specificity**.

Stock Candle prediction using CNN

PREDICT STOCK CANDLE(WHETHER IT IS A GREEN OR RED CANDLE) USING 2D AND 3D CNN

- It is my implementation of the idea of [an article](#). The GitHub page for this article is [available here](#).
- Used Python, Tensorflow, Keras, Numpy, Pandas

ISIC-2019 Melanoma Classification

CLASSIFY DERMOSCOPIIC IMAGES AMONG NINE DIFFERENT DIAGNOSTIC CATEGORIES

- Used Python, Tensorflow, Keras, Numpy, OpenCV
- I had tried different regularization techniques to overcome the overfitting problem

Certifications

- Machine Learning: [Stanford University online by Coursera](#)
- Deep Learning Specialization: [Deeplearning.ai by Coursera](#)
- AI for Medical Diagnosis: [Deeplearning.ai by Coursera](#)
- Smart Contracts: [University at Buffalo by Coursera](#)
- AWS Cloud Technical Essentials: [Amazon Web Services by Coursera](#)

language

English:	Full professional proficiency	IELTS Score - C1
German:	Beginner	